

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

SA1 –Health and Care

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INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Q.1**a) Product benefit design themes**

1. Certainty
2. Transparency
3. Prepaid or upfront payment
4. Out-of-pocket expenditure
5. Affordability
6. Simplicity
7. Flexibility

1. Certainty

In the eyes of the patient, the cost of private healthcare is highly uncertain due to:

- a) **Information asymmetry:** the severity of the healthcare cost depends on the condition diagnosed and the treatment prescribed which is not known to the patient upfront.
- b) **Tertiary healthcare:** the uncertainty of cost is exacerbated in the case of tertiary care as a typical individual would not have experienced hospitalisation frequently in their lifetime to understand the variability in healthcare cost.
- c) **High increase in tariffs:** The cost of private healthcare has increased at levels more than consumer price inflation over recent years.

Therefore, a product that introduces certainty of tertiary healthcare cost would provide peace of mind and increase the propensity to generate demand for such a product.

2. Transparency

Patients are often shocked when they receive their hospitalization bill and experience this as hidden fees being revealed at time of taking discharge from the hospital due to:

- a) A healthcare provider cannot always communicate the healthcare cost upfront. This can either be due to the provider not wanting to disclose cost for multiple service line or due to the patient's condition and treatment not being known upfront.
- b) A typical hospital patient would receive separate bills from the hospital, surgeon, anaesthetist, radiologist and pathologist.
- c) Health insurance policyholders are not always aware of the various co-payments and deductibles in their policy, and they often blame their insurer for not communicating these upfront.
- d) A policyholder may be interested in this product where the surgery or medical procedure is not covered by their policy, for example, cosmetic surgery.

Therefore, a product that can remove the "shock value" associated with medical bills and introduce transparency of cost upfront would provide peace of mind and increase the propensity to generate demand for such a product.

3. Prepaid or upfront payment

- a) Offering a product that enables pre- or upfront payment enhances the certainty experienced by the patient as his or her expectations are managed by clear communication.
- b) Prepayment can avoid the "shock value" associated with in-arrears medical billing and increase the propensity to generate demand for such a product.
- c) The prepaid feature implies that recurring premiums, contributions or payments are not required.

4. Out-of-pocket expenditure

- a) Health insurance policyholders often question the value-for-money of their cover when paying expensive premiums or contributions, while often incurring out-of-pocket expenditure due to limits being reached, co-payments or depleted medical savings account.
- b) In this instance, the product is targeted towards patients with no insurance cover, which implies the cost of this product will be paid out of the patient's pocket, but such a patient could view any higher-than-expected cost as an out-of-pocket payment such as additional fees due to medical complications, or some form of hidden fees.

Therefore, a product that can guarantee zero additional out-of-pocket expenditure – for the specific treatment required – would provide peace of mind and increase the propensity to generate demand for such a product.

5. Affordability of the target Segment

- a) A key problem is the issue of affordability. While healthcare is a basic requirement, the general population is diverse in its affordability levels. Thus, there is a primary level of demand with additional services demanded by those with higher affordability levels.
- b) This is a product that hospital can offer to their patients as an alternative to the high-cost private health insurance product with different kinds of waiting period who needs medical treatment.
- c) This product also provides an opportunity for patients without insurance cover to opt for care within the private healthcare sector.
- d) The fact that this product can be paid in cash, makes this product so much more flexible in terms of how this cost can be funded. The patient can obtain a loan, crowd-fund or pay with savings or disposable income.
- e) This product is voluntary and can be purchased on an as-and-when basis.
- f) Repeat purchasing of this product is possible, making this an innovative product feature.

6. Simplicity

- a) Health insurance products have complex benefit designs – members, policyholders and/or patients hardly have a comprehensive understanding of these products.
- b) While this product is simple in its nature: pay a once-off fee for a single procedure - in full.
- c) Cash paying patients may not necessarily be aware of the complexity of health insurance benefit design and nonetheless may be attracted to the simplicity of this product.
- d) The simplicity is further enhanced given that this product does not rely upon cross-subsidisation – a concept not always appreciated by laypersons.

7. Flexibility

- a) Health insurance products are regulated, may limiting the scope for tailored solutions.
- b) This product can be treated like an optional top-up benefit to a health insurance product, making this product a flexible solution.

(Max 15)

b) As this product is intended for patients with no health insurance coverage, affordability is critical to ensure this market will be able to afford a once-off expense in advance. The nature of the surgeries or medical procedures is therefore driven by factors affecting the affordability of the product.

1) Frequent/high demand

- a) The considered surgeries or procedures should be performed frequently by healthcare providers which translates into a direct high demand for this while frequently performed procedures also generate high volumes of granular data that is advantageous for analytical and pricing purposes.
- b) Frequently performed procedures also can stimulate efficiencies, translating into fewer complications, faster theatre times and stability in cost.
- c) Examples of frequently performed procedures include cataract, tonsil removal, appendix removal, wisdom teeth extraction, hip replacement, knee replacement, shoulder replacement, and scopes.

2) Low complication rate

- a) The considered surgeries or procedures should have low complication rates, i.e. where patients do not often require additional medical attention during the procedure while in theatre and/or during recovery.
- b) That is, the surgeries or procedures should be deemed a low-risk procedure and the probability of utilising higher levels of care is low.
- c) Surgeries or procedures with high complication rates are likely to be expensive and volatile due to unpredictably long theatre times, and recovery in expensive wards such as the Intensive Care Unit or the Intensive Critical Care units with unpredictably long lengths of stay.
- d) Surgeries or procedures with low complication rates are likely to be more affordable and have more stable costs to assist in predictable pricing and lower required margins for risk.

3) Low mortality rate

The considered surgeries or procedures should have a low mortality rate as a high mortality rate is often as a result of (or associated with) a high complication rate, which can result in expensive and unpredictable pricing.

4) Small variability in cost

The considered surgeries or procedures should have a small variability in cost to assist with the predictability in pricing.

- a) This implies that the considered surgeries or procedures should have:
- b) small variability in operation procedure.
- c) small variability in the nursing staff involved (number of nurses and speciality of nursing skills);
- d) small variability in the medication used (used during the surgery, during recovery and those dispensed to be sent home with the patient);
- e) small variability materials and consumables used during the surgery;
- f) small variability in length of stay;
- g) small variability in the level of care.

5) Short length of stay

- a) The considered surgeries or procedures should have a short length of stay, implying a fast in-hospital recovery time to assist with the affordability of the product.
- b) It could be the aim to consider day cases only where no or only one overnight stay is typically required. If the care required during recovery could be provided in the home setting this would also reduce costs and improve affordability.
- c) Preferably the in-hospital recovery should be spent in lower level of care (i.e. in the shared room or in the general ward) with a small probability of long lengths of stay in higher levels of care.

6) Sufficient supply of healthcare providers

- a) The considered surgeries or procedures should have sufficient surgeons or medical specialists available or willing to perform these procedures.
- b) The considered surgeries or procedures should be commonly done by surgeons to be considered as fairly routine surgery.

7) Other valid considerations

- a) Once-off procedures, or procedures where the likelihood of a subsequent procedure is very low.
- b) Single surgeries or common combinations of procedures that can be performed during the same admission (e.g. tonsillectomy and adenoidectomy, or breast enlargement and breast lift).
- c) Procedures of which the cost is somewhat age-agnostic.
- d) Minimal need for specialised equipment or facilities.
- e) Elective, non-emergency, and/or planned procedures that could lend itself to pre-authorisation.
- f) Procedures that are typically not covered by health insurance products.

(Max 10)

c)

1. Hospital Data

- a) Historical claims data should be relevant, reliable and credible. If data for the calendar years of 2020 and 2021 should be used with suitable adjustments for the COVID-19 pandemic claims.
- b) The data should be used to ascertain the severity of the procedure cost only, while there is no need to estimate utilisation trends or claims incidence rates. We know with certainty or very high probability that this procedure will be performed upon sale (i.e. the probability of occurrence is 100%).
- c) The hospital data should provide historic cost of medical treatment or cost for these procedures, medicines, consumables and other costs including:
 - Salary of staff
 - Administrative expenses
 - Medical Support Services
 - Repairs and replacements
 - Operational costs
 - Profit.
- d) The above investigation could be performed by means of an activity-based costing exercise.
- e) Identify the relevant ICD / CPT codes for each conditions /procedure while liaising with a healthcare professional or a clinical coding specialist to understand the implications of including or excluding certain CPT codes.
- f) Apply rules to the data as per protocols or clinical entry criteria, for example
- g) Age limits: is the procedure for children only, or adults only.
- h) Length of stay limitations: if the intention is to consider day care cases only, then exclude data points where the length of stay exceeds 1 day or procedures with a high proportion of cases with more than 1 day length of stay. In case Day care procedures are not covered, suitable adjustments are required accordingly.

- i) Level of care: if the intention is to consider cases with limited probability of higher levels of care, then exclude data points where patients required care in High Care or ICU wards or procedures where patients often required care in High Care or ICU wards.
- j) Consider and treat outliers suitably.
- k) Perform a scenario test where the above data points are excluded, and another scenario where no data points are excluded, and use the relative difference of these scenarios to quantify the risk associated with complications.
- l) Investigate the data whether the hospital typically owns the equipment required to perform the procedure, or does the specialist need to bring the required equipment and quantify the difference of the usage fee between the hospital and the specialist.
- m) Investigate whether the data includes cases where the procedure can be done with local anaesthesia, or general anaesthesia only.
- n) Investigate whether the data reveals common procedure combinations, e.g. colonoscopy and gastroscopy that should be grouped as one product...
- o) ...and assess whether the combined cost can be less than the sum of the individual procedures.
- p) Investigate whether the data reveals if there are multiple procedures per patient and whether there are cost differences in these cases, e.g. if one patient requires two cataract removals or multiple teeth extractions.

2. Protocols

Incorporate the relevant protocols of this product in the pricing, for example:

- a) The bill of the materials and consumables can be agreed upfront and priced for as the sum of each material, assuming the healthcare professionals will abide by these prescribed materials.
- b) The maximum length of stay can be agreed upon and priced accordingly
- c) Protocols (like the above) can also be informed by the historic data, while liaising with a healthcare professional to appreciate the clinical nuances.

3. Provider networks

Set up a network of efficient providers and negotiate favourable rates to assist in affordability of the overall product:

- a) Specialists
- b) Anaesthetist
- c) Pathology labs
- d) Radiology.

The negotiation strategy should include that these products enable a new market to be channelled to these providers.

4. Determine the healthcare cost

The healthcare cost per procedure would represent the sum of the following:

- a) Hospital cost
- b) Medication cost
- c) Specialist fee
- d) Anaesthetist fee
- e) Pathology fee
- f) Radiology fee.

The healthcare cost should be sufficient to cover all the fees considering future medical inflation and there is no cross-subsidisation mechanism.

5. Expenses

The following expenses need to be added to the healthcare cost

- Administration charges
- Margin for adverse deviation
- Profit margin.

6. Tax

GST needs to be added to the final price.

7. Competitive Comparison

- a) A competitive comparison would be difficult to perform as similar products are unlikely to be available in the market.
- b) And if there are comparable products in the market, it is unlikely that the cost breakdown would be publicly available.

(Max 15)

[40 Marks]

Q.2)

a) Factors to consider

- 1) Type of contract – Facultative, Treaty
- 2) Type of cover – Quota share, Surplus, Per risk excess of loss
- 3) Basis of Cover -Loss occurring, Risk attaching claims made
- 4) Line of Business
- 5) Data:
 - Homogeneous,
 - credible,
 - relevant,
 - reliable
 - complete and accurate
- 6) Length of Origin and development period
- 7) Claim type
- 8) Cedent type

(2)

b) Additional Data Issues

- 1) Claims reporting delays are longer for reinsurers as insurers generally submits bordereau after end of quarter.
- 2) There is greater heterogeneity of exposure for reinsurance data. This may be due to more contract types, differing terms and conditions or case reserving philosophies. This is an issue since most reserving methods require large volumes of homogenous data.
- 3) Industry benchmarks may be less applicable. This follows from the heterogeneity of exposures which may lead to very different claims development patterns.
- 4) The information received about losses is less detailed than the information insurers receive. This is because the information is provided on an aggregated basis.

- 5) Because of the complexity and individuality of reinsurance risks, it is difficult for reinsurers to have IT systems that capture perfectly all the contracts written and their key features.
- 6) There is a greater tendency for claims to develop upwards for non-proportional reinsurance, particularly for liability business.
- 7) Due to the heterogeneity of reinsurers' exposures, different reinsurers can experience very different claims development patterns, which makes industry wide benchmarks potentially less appropriate.

(2)

c) Difference

- 1) Using an accident year cohort the chain ladder will estimate ultimate claims for claims occurring in the given accident year. The estimated ultimate claims only include claims relating to expired risk.
- 2) Using an underwriting year cohort the chain ladder will estimate ultimate claims for claims arising from policies sold in the given underwriting year. The estimated ultimate claims include claims relating to expired and unexpired risks.

(2)

d) Reason

- 1) IBNR reserves relate to expired exposure i.e. the claim event has already occurred. This is consistent with earned premium.
- 2) As at the valuation date a portion of the written premium will be unearned. This unearned portion will cover claim events occurring after the valuation date. Using written premium will thus overstate the IBNR

(2)

e) IBNR calculation

Percentage of Earned at 31/12/2025	Earned Premium ₹ In Lakhs	Ultimate Claims as at 31/12/2025 ₹ In Lakhs	Ultimate Loss Ratio	Ultimate Claims incurred ₹ In Lakhs	IBNR ₹ In Lakhs
100%	1,200	1,111.16	93%	1,111.16	166.76
66.67%	1,250	1,661.54	89%	1,107.69	27.69
33.33%	1,100	2,500.00	76%	833.33	83.33
Total					277.79

(5)

f) Issues to consider

- 1) If some Critical Illness policies have a significantly different term than three years it could distort the earned premium calculation, and hence the IBNR reserve.
- 2) The claims development percentages were derived for Critical Illness business written locally in India, which may not be applicable to Critical illness business reinsured in home country of your company.
- 3) The IBNR calculated is gross of any retrocession. The retrocession recoveries will need to be calculated to determine the net of retrocession IBNR.

- 4) The level of margin that should be booked. This will depend on the company's reserving policy.
- 5) Whether there were any changes to the claims' development patterns which may invalidate the assumption that the past is a good guide to the future.
- 6) If the assumption that the risk is earned uniformly, which is needed when calculating the earned premium, is appropriate.
- 7) If any additional assumptions need to be made for direct or indirect claims expenses.

(2)

[15 Marks]**Q. 3)****a) How income protection (IP) meets the needs of customers**

- 1) IP provides the insured with an income during periods of short or long-term disability, sickness or accident (referred to as incapacity).
- 2) Its function is to replace part of the income lost as a result of the incapacity.
- 3) The income lost would also be a major problem for any dependants the insured may have, eg children and spouse.
- 4) The insured can claim any number of times during the term of the policy, subject to the policy conditions.
- 5) On individual contracts, premiums are unlikely to increase with age, ...
- 6) ... but may increase with inflation and may be reviewable depending on experience.
- 7) There are, however, limitations on the benefits, ...
- 8) ... for example in the form of deferred periods, and in the definition of disability.
- 9) The product may also be used to cover loan repayments (including the vehicle used to repay the loan) often called Mortgage Payment Protection Insurance.
- 10) Unemployment may also be a covered peril.
- 11) The benefits from an IP policy may also be used to cover insurance premium payments on other insurance policies, eg pension contributions.
- 12) Locum Protection Insurance is another form of IP.
- 13) This is where professionals (eg a group of doctors in one practice) use the benefits to pay for temporary employment of a replacement for one of their colleagues in the event of that colleague's incapacity.
- 14) The customers of an insurer include employers.
- 15) Group versions of IP policies are available for purchase by employers.
- 16) For employees unable to work, the employer will at least have to pay statutory sickness benefits. It may also have to pay for a replacement employee.
- 17) IP policies will protect against some of this loss and they encourage a return to work and so may help to minimise these costs.
- 18) The employer also has a moral obligation to keep its workforce happy. IP is a benefit often valued by the workforce, which may also encourage staff to work for, and remain with, the company.
- 19) It is particularly keen to keep the senior key staff happy, as they will be more valuable to the company's future, and are more difficult to replace.
- 20) IP policies will provide disability income and will also encourage a return to work via the use of benefits limited to a proportion of pre-disability income.
- 21) IP products may provide a partial benefit to those who return to work on a part-time basis or in a less strenuous role.
- 22) This will benefit the employer (by encouraging a return to work) and the employee (by helping with their recovery).
- 23) The employer may also get a tax advantage of providing IP benefits.
- 24) Premiums paid by the employer qualify for tax relief, and the benefits are treated as a business expense.
- 25) The aim for the employer will be an overall cost outgo less than the benefits provided (both tangible and perceived) to all the employees.
- 26) The employer will not want the administration of the IP policies to be a burden.
- 27) The assessment of claims and the provision of benefits needs to be straightforward.

- 28) Subject to keeping all levels of staff happy, the employer wants the benefits to employees to be as consistent as possible and so be seen to be fair.
- 29) This would mean offering the policies to all staff, and perhaps offering a benefit consistent with salary.
- 30) The customers of an insurer include the sales staff.
- 31) The distributors of the contracts will need policies that are well-designed to meet
- 32) customers' needs and with clear policy wording.
- 33) This is important since IP policies can often be complex.
- 34) A popular product will encourage cross-selling, since the sales staff would be more willing to promote other products from the same company.
- 35) The premiums and benefits should be competitive and good value for money, to ensure marketability.
- 36) The product should be reasonably flexible, giving the desired benefits. The policy could be packaged with other benefits such as PMI.
- 37) The remuneration terms (eg commission) should be satisfactory to both the distributors and the company.
- 38) The underwriting and claims process should be reasonably straightforward, not too onerous and fair to all parties so that the distributor can offer good service.
- 39) Any sales regulations requirements should be met by the product's design and packaging. For example, any literature should conform to guidelines.

(Max 15)

b) Why monitor?

- 1) Given that there are substantial volumes of business the company needs to be aware of how the business is performing.
- 2) This is so that it can spot possible trends in order to take early action.
- 3) The key reason for monitoring is to identify divergences in actual experience from that which was assumed in the contract pricing, as part of the actuarial control cycle for the business
- 4) The company is particularly interested in whether the contract is proving as profitable as expected ...
- 5) ... and is giving shareholders an acceptable return on their capital.
- 6) If not, the company can consider what it needs to change for existing business (to the extent it can) and for new business.
- 7) On its in-force IP business it might be able to improve its claims underwriting if that is found to be a problem.
- 8) There is little it can do on pricing if terms are guaranteed.
- 9) Either way, the company can revise assumptions in its embedded value calculations, to maintain an accurate view of the worth of the business and to recognise profits (and losses) appropriately.
- 10) If the existing contracts are reviewable, then the results can be used to change either the premiums or the benefits given.
- 11) However, the scope to do this will to an extent be limited by competitive forces and the need to treat customers fairly.
- 12) For new business it will want to use the results of monitoring to help it re-price, where this is shown to be necessary.
- 13) The information gained can be used to revise assumptions and produce new premiums.
- 14) By monitoring business volumes and mix the company can also gain some information on the marketability and competitiveness of the product by sector.
- 15) Monitoring mix is especially important if (as is likely) there is cross-subsidy between policies.
- 16) The company also needs to monitor aspects of experience such as claim inceptions and terminations, expenses and persistency in order to check the appropriateness of supervisory valuation assumptions and for PRA reporting.

(Max 8)

c) Methods

- 1) The company may use an analysis of surplus.
- 2) This will help to identify if any items of experience are drifting away from assumptions.
- 3) Alternatively, a model office approach could be used.
- 4) The company will also wish to carry out investigations of the key individual items of experience.
- 5) For IP this will mean morbidity (split between inception and recovery), lapses and expenses.
- 6) For each analysis, group and individual business may be treated separately as they normally demonstrate different characteristics.
- 7) Morbidity would be investigated by studying experience over a period of, say, 3–5 years, splitting data at least by:
 - gender
 - age-group
 - occupational class
 - deferred period
- 8) ... and possibly by smoker status and region.
- 9) It would be essential to study claim inception and termination rates separately, as the necessary action would differ depending on where any problems emerged.
- 10) Will also split claim termination data by reason (death or recovery) and duration.
- 11) The large volumes should allow a reasonably large number of cells in the analysis,
- 12) while retaining credibility.
- 13) The inception and termination rates calculated would be compared with pricing
- 14) assumptions.
- 15) The company may try to identify trends, but even with large volumes this may not be possible, particularly for terminations where data is split further.
- 16) Expenses would probably be investigated as part of regular company-wide expense investigations.
- 17) These could classify non-commission expenses as per premium, per sum assured and per policy.
- 18) A split into initial, renewal, termination and investment expenses is also likely.
- 19) Any overheads that were not clearly identified as relating to products would have to be split by pragmatic approaches, so that IP were allocated an appropriate share.
- 20) For sensible comparisons over time, adjustment for inflation would be necessary.
- 21) Lapses on the IP portfolio would be analysed by duration in force and by distribution channel (if more than one).
- 22) Other splits such as gender and age, level of benefit, payment frequency and method may be considered if there is sufficient data.
- 23) Would like to monitor by individual salesperson or broker to see if any individual sales outlets are a problem.
- 24) Care would be necessary in using lapse results for future pricing, because they can be quite volatile, especially in response to economic conditions.
- 25) The company may also monitor its underwriting and reinsurance limits, to ensure that these are still at appropriate levels for the developing portfolio.
- 26) We should also monitor economic factors and competition, to assess the possible effects of such external influences on our future experience.

(Max 10)

d) Suitable assets

- 1) The company should invest so as to maximise return, subject to acceptable risk.
- 2) The benefit payments from the IP policies may consist of non-increasing benefits, or, perhaps more likely (particularly for group benefits), benefits that increase in line with
- 3) salary inflation (perhaps limited to an absolute amount).
- 4) We therefore require further information about the mix of business and how benefits increase, both during the policy term and during claim payment.
- 5) Expenses can also be considered to increase in line with an index, again perhaps salary inflation.
- 6) Premiums will normally be level, but again may increase over time in accordance with the contract wording. These can be netted off the increasing benefit where applicable.

- 7) Again, more information on premiums will be needed from the company.
- 8) Assets should be in the same currency as that of the liabilities. We need to know
- 9) whether any of the policies cover overseas policyholders or employees.
- 10) If there are liabilities in other countries, these could be matched with overseas investments.
- 11) For the benefits guaranteed in money terms, a mixture of suitable short-term investments would be a suitable match.
- 12) For example, a variety of fixed-interest bonds and cash.
- 13) For the increasing benefits, assets that increase similarly, such as index-linked gilts, would be suitable.
- 14) An alternative would be other assets expected to give a real return, such as equities or property, although the volatility and marketability of these would need to be taken into account
- 15) We should therefore like to know the size of the insurer, and its level of investment expertise to see whether these alternatives would suit.
- 16) A small company with minimal expertise would probably not want a large proportion of volatile assets.
- 17) The size of the insurer's free surplus will also have a big influence on the desirability of more risky assets, since a well-financed company could afford a more mismatched and risky investment approach.
- 18) As the company is a proprietary one, it has a duty to its shareholders to sustain and enhance the return it achieves on its assets backing the free surplus. This would imply some investment in equities and property.
- 19) We should also like to know what legislative restrictions there are on the investments we can make.
- 20) As the company is growing rapidly, provided it remains solvent, it can use incoming premiums to pay claims. [$\frac{1}{2}$]
- 21) However, future premium income is not guaranteed and so caution should be exercised.
- 22) We should like to know the recent claims experience of the insurer. If the outgo is highly variable, a more cautious approach should be taken, retaining a high proportion of liquid assets.
- 23) Also, the extent of guarantees should be known, as a large proportion of guaranteed business may make net outgo more volatile.
- 24) Short-dated gilts and cash could be held to back the capital requirements. This would reduce the exposure to market risk and therefore limit the size of the Solvency Capital Requirement .
- 25) The need to minimise capital requirements is less important to a company with a large free surplus, so we should like to know the size of the SCR in comparison to the size of the free surplus.
- 26) We also need to know the tax position of the company, since this will also influence the asset mix.
- 27) The results of any ALM exercise would be useful.

(Max 12)

[45 Marks]
